

HCF and LCM

LCM (Least common multiple) of two or more given numbers is the least number that is exactly divisible by each of them.

HCF (Highest common factor) of two or more numbers is the greatest number that divides each of them exactly. HCF is also known as the 'Highest common Divisor' (HCD) and the Greatest Common Measure (GCM).

The concept of multiples and factors

→ If X, Y, and Z are three natural numbers and $X \times Y = Z$, then

- X and Y are called the factors of Z.
- Z is said to be divisible by X and Y.
- Z is said to be a multiple of X and Y.

Example: The set of positive Integers which are factors of 18 is (1, 2, 3, 6, 9, 18).

Basic Concepts of H.C.F. and LCM

Method of finding H.C.F.

To find the HCF of the given numbers

- Break the given numbers into their prime factors.
- The HCF will be the product of all the prime factors common to all the numbers.

Let us learn the process of finding HCF with the help of some solved examples.

Example:-

Find the HCF of 96, 36 and 18 ?

Solution:-

$$96 = 2 \times 3 \times 2 \times 2 \times 2 \times 2$$

$$36 = 2 \times 3 \times 2 \times 3$$

$$18 = 2 \times 3 \times 3$$

Therefore, the HCF of 96, 36 and 18 is the product of the highest number of common factors in the given numbers i.e., $2 \times 3 = 6$. In other words, 6 is the largest possible integer, which can divide 96, 36 and 18 without leaving any remainder.

Example:- Find the H.C.F. of 42 and 70 ?

Solution:- $42 = 3 \times 2 \times 7$

$$70 = 5 \times 2 \times 7$$

Hence, H.C.F. of 42 and 70 = $2 \times 7 = 14$.

HCF by Division Method

Example:- HCF of 24, 48, 72, and 100.

Solution:- To start the division method select the smallest two numbers./

$$\begin{array}{r} 24 \overline{)48} \begin{array}{l} 2 \\ \times X \\ \hline 48 \\ \hline 0 \end{array} \quad \begin{array}{r} 48 \overline{)72} \begin{array}{l} 1 \\ \times X \\ \hline 48 \\ \hline 24 \end{array} \quad \begin{array}{r} 24 \overline{)100} \begin{array}{l} 4 \\ \times X \\ \hline 96 \\ \hline 4 \end{array} \end{array}$$

$$\text{HCF of 24 and 48} = 24$$

$$\text{HCF of 24, 48 and 72} = 24$$

$$\text{HCF of 24, 48, 72, and 100} = 4.$$

Example:- HCF of 1785, 1995, 3381.

Solution:-

$$\begin{array}{r} 1785 \overline{)1995} \begin{array}{l} 1 \\ \times 1785 \\ \hline 1785 \\ \hline 210 \end{array} \quad \begin{array}{r} 105 \overline{)3381} \begin{array}{l} 32 \\ \times 105 \\ \hline 315 \\ \hline 231 \end{array} \end{array}$$

$$\text{HCF of 1785 and 1995} = 105$$

$$\text{HCF of 1785, 1995 and 3381} = 21$$

NOTE :-

- HCF of two prime numbers is always 1.
- HCF of co-prime numbers is always 1.

Method of finding L.C.M

The Least Common Multiple of two or more numbers is the smallest number which is exactly divisible by all of them. In other words, it is the product of the highest powers of all the prime factors of the given numbers.

To find the LCM of given numbers:

- Break the given numbers into their prime factors.
- The LCM will be the product of the highest power of all the factors that occur in the given numbers.

Let us take some solved examples.

Example:- Find the LCM of 96, 36 and 18.

$$\begin{aligned} \text{Solution: } 96 &= 2 \times 2 \times 2 \times 2 \times 2 \times 3 \\ &= 2^5 \times 3^1; 36 = 2 \times 2 \times 3 \times 3 = 2^2 \times 3^2 \\ 18 &= 2 \times 3 \times 3 = 2^1 \times 3^2 \end{aligned}$$

Therefore, LCM of 96, 36 and 18 is the product of the highest powers of all the prime factors, i.e. $2^5 \times 3^2 = 32 \times 9 = 288$. That is, 288 is the smallest integer which is divisible by 96, 36 and 18 without leaving any remainder.

Example:- Find the LCM of 42 and 70

Solution:- $42 = 3 \times 2 \times 7$

$$70 = 5 \times 2 \times 7$$

Hence, LCM is $2 \times 3 \times 5 \times 7 = 210$.

Example:- LCM of 6, 12, 8 ?

Solution:-

$$\begin{array}{r} 2 \overline{)6, 12, 8} \\ 2 \overline{)3, 6, 4} \\ 3 \overline{)3, 3, 2} \\ 2 \overline{)1, 1, 2} \\ 1 \overline{)1, 1, 1} \end{array}$$

$$\text{LCM} = 2 \times 2 \times 3 \times 2 = 24$$

HCF of 6, 12, 18

Firstly find out the factors of 6, 12, 18 and then multiply the common factors.

$$\begin{array}{r} 2 \overline{)6} \begin{array}{l} 3 \\ \times 2 \\ \hline 6 \\ \hline 0 \end{array} \quad \begin{array}{r} 2 \overline{)12} \begin{array}{l} 6 \\ \times 2 \\ \hline 12 \\ \hline 0 \end{array} \quad \begin{array}{r} 2 \overline{)18} \begin{array}{l} 9 \\ \times 2 \\ \hline 18 \\ \hline 0 \end{array} \end{array}$$

$$6 = 2 \times 3, 12 = 2 \times 2 \times 3, 18 = 2 \times 3 \times 3$$

$$\text{HCF} = 2 \times 3 = 6$$

Try finding HCF and LCM of 3, 6, 9, 12 yourself. HCF of 3, 6, 9, 12 can also be found by Division method. It is useful when the numbers are bigger.

NOTE :-

- HCF of A, B and C is the highest divisor which can exactly divide A, B, and C.
- LCM of A, B and C is the lowest dividend which is exactly divisible by A, B, and C.

There is one very important relationship, given below, between two numbers and their HCF and LCM. Many problems have appeared in various competitive exams based on this relationship.

Important Concepts:-

(1).

$$\text{LCM} \times \text{HCF} = 1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}$$

Example:- For numbers 8 and 12,

$$\text{LCM} = 24 \text{ and } \text{HCF} = 4$$

$$\text{Now, } \text{LCM} \times \text{HCF} = 24 \times 4 = 96$$

$$\text{also, } 8 \times 12 = 96$$

(2). HCF of some numbers is always a factor of LCM of the numbers.

(3).

$$\text{LCM of Fraction} = \frac{\text{LCM of numerator}}{\text{HCF of denominator}}$$

(4).

$$\text{HCF of Fraction} = \frac{\text{HCF of numerator}}{\text{LCM of denominator}}$$

Example: LCM and HCF of $\frac{1}{2}$, $\frac{2}{3}$ and $\frac{3}{4}$

$$\begin{aligned} \text{Solution:- } \text{LCM} &= \frac{\text{LCM of numerator}}{\text{HCF of denominator}} \\ &= \frac{\text{LCM of } 1, 2, 3}{\text{HCF of } 2, 3, 4} = \frac{6}{1} \end{aligned}$$

$$\begin{aligned} \text{HCF} &= \frac{\text{HCF of numerator}}{\text{LCM of denominator}} \\ &= \frac{\text{HCF of } 1, 2, 3}{\text{LCM of } 2, 3, 4} = \frac{1}{12} \end{aligned}$$

(5) Co- Prime numbers :

If the HCF of two numbers is 1 then they are called co-prime numbers.

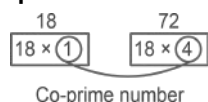
$$(6) \frac{\text{LCM}}{\text{HCF}} = \text{Product};$$

where LCM and HCF are of two numbers N_1 and N_2 . If we find two co-prime factors, F_1 and F_2 , of the Product as obtained above.

(7)



Example:-



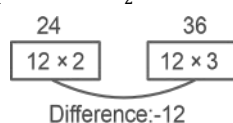
(8).

If 1st number = N_1 and 2nd number = N_2 HCF of $N_1 = Hx$ HCF of $N_2 = Hy$ And HCF of N_1 and $N_2 = H$

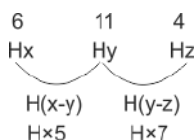
So,

Difference between N_1 and $N_2 = Hx - Hy$ $= H(x - y)$

Example :-

 $N_1 = 24$ and $N_2 = 36$ 

Note :- HCF is always either the difference of two numbers or factors of difference of two numbers.



(9). When the second divisor is a factor of the first divisor, then the second remainder is obtained by dividing the first remainder by the second divisor.

Example :- When 29 is divided by 8, the remainder obtained is 5, then what will be the remainder when the same number is divided by 4?

Solution:- Here, the second divisor that is 4 is a factor of the first divisor that is 8. So, on dividing the first remainder that is 5 by the second divisor that is 4, we get our second remainder which is 1. So, the required answer is 1.

Variety Questions

Q.1. What is the sum of all two-digit numbers divisible by both 3 and 5?
 Graduate Level 24/07/2025 (Shift - 3)
 (a) 300 (b) 315 (c) 330 (d) 360

Q.2. Let $a = 2^3 \cdot 3^2 \cdot 7$ and $b = 2^2 \cdot 3 \cdot 5^2$. Which of the following is true?
 Higher Secondary 29/07/2025 (Shift - 2)
 (a) $L.C.M(a, b) = 2^3 \cdot 3^2 \cdot 5^2 \cdot 7$
 (b) $G.C.D(a, b) = 2^3 \cdot 3^2$
 (c) $L.C.M(a, b) \times G.C.D(a, b) = a + b$
 (d) $L.C.M(a, b) = G.C.D(a, b)$

Q.3. The LCM of $x^2 - 8x + 15$ and $x^2 - 5x + 6$

is:

- (a) $(x + 5)(x + 2)(x + 3)$
 (b) $(x - 5)(x - 2)(x - 3)$
 (c) $(x + 5)(x - 2)(x - 3)$
 (d) $(x - 2)(x - 3)^2(x - 5)$

Q.4. The greatest possible length that can be used to measure exactly the lengths of 3 m 15 cm, 5 m, and 6 m 85 cm is:

- (a) 11 cm (b) 7 cm (c) 9 cm (d) 5 cm

Q.5. What is the HCF of $(x^6 + 1)$ and $(x^4 - 1)$

- (a) $(1 + x^2)$ (b) $(1 + x)$ (c) 1 (d) $(1 - x^2)$

Q.6. The LCM of two prime numbers x and y ($x > y$) is 533. The value of $4y - x$ is:

- (a) 11 (b) 21 (c) 18 (d) 23

Q.7. What is the HCF of the polynomials $(x^3 - 8)$, $(x^3 - 6x^2 + 12x - 8)$ and $(x^3 - 4x^2 + 4x)$?

- (a) $(x - 1)$ (b) $(x - 2)$ (c) $(x - 8)$ (d) $(x - 4)$

Q.8. Radha, Pratima and Reena begin to jog around a circular path and they complete their revolutions in 50 seconds, 75 seconds and 100 seconds, respectively. After how much time (in minutes) will they meet together at the starting point for the first time?

- (a) 6 (b) 5 (c) 4 (d) 3

Q.9. Sanjay wants to exactly measure the lengths 7000 mm, 3850 mm and 12950 mm using a single measuring tape. What will be the greatest possible length (in cm) of the measuring tape?

- (a) 40 (b) 45 (c) 35 (d) 30

Q.10. Two tankers contain 850 litres and 680 litres of oil. Find the maximum capacity of a container which can measure the oil of both tankers when used, an exact number of times.

- (a) 425 litres (b) 680 litres
 (c) 340 litres (d) 170 litres

Q.11. Find the greatest two digit number which on dividing 219, 365, 511 leaves the remainders 3, 5, 7, respectively.

- (a) 63 (b) 82 (c) 53 (d) 72

Q.12. Five bells commence tolling together and toll at intervals of 5, 10, 15, 20 and 25 seconds respectively. In 1 hour 20 minutes, how many times do they toll together?

- (a) 20 (b) 15 (c) 14 (d) 17

Q.13. There is a circular path around a sports field. Rahul takes 15 minutes to drive one round of the field, while Anil takes 18 minutes for the same. Suppose they both start from the same point and at the same time, and go in the same

direction. After how many minutes will they meet again at the starting point?

- (a) 120 (b) 100 (c) 80 (d) 90

Q.14. Let $x = 224$ and, $y = 322$. If the highest common factor of $23x$ and $a \times y$ is divisible by x and y , then what can be the possible value of a ?

- (a) 16 (b) 8 (c) 12 (d) 4

Q.15. The highest common factor of 108, 72 and $5a$ is a . What can be the least common multiple of 108, 72 and a ?

- (a) 432 (b) 324 (c) 108 (d) 216

Q.16. What is the greatest positive integer that divides 554, 714 and 213 leaving the remainder 43, 57 and 67, respectively?

- (a) 95 (b) 71 (c) 83 (d) 73

Q.17. Find the HCF of

$(4^{315} - 1)$ and $(4^{25} - 1)$.

- (a) 1 (b) $(4^{25} - 1)$ (c) 1024 (d) 1023

Q.18. 120 apples, 240 oranges and 150 pears are packed in cartons in such a way that each carton has the same number of fruits, each carton contains only one type of fruit and fruit is left unpacked. What is the smallest possible number of cartons needed for the purpose?

- (a) 50 (b) 40 (c) 17 (d) 30

Q.19. The traffic lights at 3 different road crossings change after every 48 sec, 72 sec and 108 sec, respectively. If they all change simultaneously at 8 : 20 a.m., then at what will they next change again simultaneously?

- (a) 8 : 27 : 12 a.m. (b) 8 : 33 : 32 a.m.
 (c) 8 : 12 : 18 a.m. (d) 8 : 40 : 14 a.m.

Q.20. A and B are two prime numbers such that $A > B$ and their LCM is 209. The value of $A^2 - B$ is?

- (a) 350 (b) 372 (c) 361 (d) 339

Q.21. 13, a , b , c are four distinct numbers and the HCF of each pair of numbers (13, a) : (13, b) : (13, c) is 13, where a , b , c are each less than 60 and $a < b < c$. What is the value of $\frac{a+c}{b}$?

- (a) 3.5 (b) 2 (c) 5 (d) 4.5

Q.22. The sum of two numbers is 1215 and their HCF is 81. If the numbers lie between 500 and 700, then the sum of reciprocals of the numbers is

- (a) $\frac{5}{1512}$ (b) $\frac{5}{378}$ (c) $\frac{5}{702}$ (d) $\frac{5}{1188}$

Q.23. If r is the remainder when each of 6454, 7306 and 8797 is divided by the



greatest number d ($d > 1$), then $(d - r)$ is:

- (a) 126 (b) 64 (c) 137 (d) 149

Q.24. In finding the HCF of two numbers by division method, the quotients are 1, 8 and 2 respectively, and the last divisor is 105, what is the sum of the numbers?

- (a) 3570 (b) 3885 (c) 3780 (d) 3675

Q.25. When the smallest number x is divided by 5, 6, 8, 9 and 12, it gives remainder 1 in each case. But x is divisible by 13. What will be the remainder when x will be divided by 31

- (a) 1 (b) 5 (c) 3 (d) 0

Q.26. The Highest Common Factor and Lowest Common Multiple of two numbers p and q are A and B respectively. If $A + B = p + q$, then the value of $A^3 + B^3$ is:

- (a) p^3 (b) q^3 (c) $p^3 + q^3$ (d) $p^3 - q^3$

Q.27. What is the largest number that divides 460, 491 and 553 and leaves the remainder 26 in each case?

- (a) 27 (b) 35 (c) 33 (d) 31

Q.28. A is the smallest three-digit number which when divided by 3, 4 and 5 gives remainder 1, 2 and 3 respectively. What is the sum of the digits of A ?

- (a) 11 (b) 10 (c) 6 (d) 8

Q.29. The product of two numbers is 6760 and their HCF is 13. How many such pairs of numbers can be formed?

- (a) 2 (b) 3 (c) 1 (d) 4

Q.30. An oil merchant has 3 varieties of oil of volumes 432, 594 and 702 respectively. The number of cans of equal size that would be required to fill the oil separately is:

- (a) 13, 15, 17 (b) 8, 11, 13
(c) 8, 13, 15 (d) 6, 9, 11

Q.31. Which of the following statements is true?

- (a) LCM of two natural numbers is divisible by their HCF.
(b) $HCF + LCM$ of two numbers = Product of the two numbers.
(c) Two prime numbers are co-prime numbers if their LCM is 1.
(d) HCF of two numbers is the smallest common divisor of both numbers.

Q.32. How many numbers between 400 and 700 are divisible by 5, 6 and 7?

- (a) 2 (b) 5 (c) 10 (d) 20

Q.33. If r is the remainder when each of 4749, 5601 and 7092 is divided by the greatest possible number d ($d > 1$), then the value of $(d + r)$ will be:

- (a) 276 (b) 271 (c) 298 (d) 282

Q.34. Let x be the least 4-digit number which when divided by 2, 3, 4, 5, 6 and 7 leaves a remainder of 1 in each case. If x lies between 2800 and 3000, then what is the sum of digits of x ?

- (a) 15 (b) 16 (c) 12 (d) 13

Q.35. Let x be the least number which when divided by 15, 18, 20 and 27, the remainder in each case is 10 and x is a multiple of 31. What least number should be added to x to make it a perfect square?

- (a) 39 (b) 37 (c) 43 (d) 36

Q.36. When 12, 16, 18, 20 and 25 divide the least number x , the remainder in each case is 4 but x is divisible by 7. What is the digit at the thousands' place in x ?

- (a) 5 (b) 8 (c) 4 (d) 3

Q.37. When 6892, 7105 and 7531 are divided by the greatest number x , then the remainder in each case is y . What is the value of $(x - y)$?

- (a) 123 (b) 137 (c) 147 (d) 113

Q.38. Let x be the greatest number which when divides 6475, 4984 and 4132, the remainder in each case is the same. What is the sum of digits of x ?

- (a) 4 (b) 7 (c) 5 (d) 6

Examination wise Questions

SSC CGL 2025 Tier - 1

Q.39. What is the least square number that is divisible by 6, 9, and 15?

SSC CGL 26/09/2025 (Shift 1)

- (a) 1800 (b) 900 (c) 400 (d) 1600

Q.40. What is the least square number that is exactly divisible by 3, 5, 12, and 15?

SSC CGL 14/10/2025 (Shift 2)

- (a) 3600 (b) 2304 (c) 3364 (d) 4489

SSC Selection Post (XIII)

Q.41. Raman and Rakesh start jogging in a park at the same time and from the same point. Raman takes 10 minutes to complete one round, and Rakesh takes 15 minutes. After how many minutes will they meet again at the starting point

Higher Secondary 29/08/2025 (Shift - 2)

- (a) 30 minute (b) 40 minute

- (c) 60 minute (d) 80 minute

Q.42. Find the least number which leaves remainder 4 when divided by 6, 9, and 10. Graduate Level 29/07/2025 (Shift - 3)

- (a) 90 (b) 94 (c) 85 (d) 75

Q.43. What is the smallest number that, when divided by 3, 4, 5, 6, and 8 leaves the remainder 4?

Graduate Level 29/08/2025 (Shift - 1)

- (a) 124 (b) 120 (c) 128 (d) 240

eduquity-based pattern (ebp) Questions

Q.44. A milk vendor has 75 litres of cow milk, 45 litres of toned milk and 30 litres of full cream milk. These should be placed in different trays such that each tray contains the same quantity of milk and two kinds of milk should not be mixed in a tray. What is the least number of trays required to place all the milk?

- (a) 15 (b) 25 (c) 10 (d) 20

Q.45. Three numbers are in the ratio 5 : 7 : 9, and their LCM is 1260. What is the difference between the largest and the smallest number?

- (a) 24 (b) 20 (c) 16 (d) 32

Q.46. Three red lights on the same road start blinking at the same time and blink repeatedly after every 4, 6 and 8 minutes, respectively. How many times will they blink together in an interval of 3 hours, excluding blinking together at the starting moment?

- (a) 5 (b) 7 (c) 6 (d) 4

Q.47. As part of the partial work from home process, Amita went to office every 2nd day, Amrita went to office every 3rd day, and Tosita went to office every 6th day, respectively. All three went to office on 31 October 2021. On how many days did all three attend office on the same date during the remaining two months of the year?

- (a) 9 (b) 10 (c) 11 (d) 12

Q.48. Find the least number which when divided by 20, 25, 35 and 40 leaves remainders 14, 19, 29 and 34, respectively.

- (a) 1284 (b) 1294 (c) 1384 (d) 1394

Q.49. If two positive integers x and y are written as $x = a^2b^3$ and $y = a^3b$; and a and b are prime numbers, then HCF (x, y) is:

- (a) ab^2 (b) a^2b^2 (c) ab (d) a^2b

Q.50. x, y and z are divisible by 2, 3 and 4, respectively. What is the highest common factor (HCF) of $6x, 4y$ and $3z$?

- (a) At least 36 (b) At least 12
(c) At least 18 (d) At least 16

Q.51. Find the HCF of:

$$3^3 \times 11^2, 2^4 \times 3^2 \times 5^3 \times 11^3 \text{ and } 3 \times 5 \times$$



7×11

(a) 385 (b) 1815 (c) 33 (d) 165

Q.52. Two numbers are in the ratio of 7 : 9 and their LCM is 630. Find the numbers
(a) 56, 72 (b) 77, 99 (c) 70, 90 (d) 63, 81

Q.53. If the HCF of 156 and 96 is expressible in the form of $156 \times 12 + 96A$, then the value of A is:

(a) -19.735 (b) 19.735
(c) -19.375 (d) 19.375

Q.54. The HCF of 1.75, 5.6 and 7 is:
(a) 0.35 (b) 3.5 (c) 0.07 (d) 0.7

Q.55. The LCM of $x^3 - 27$, $x^2 - x - 6$, $x + 2$ is:

(a) $(x - 3)(x + 2)(x^2 - 3x + 9)$
(b) $(x + 3)(x + 2)(x^2 - 3x + 9)$
(c) $(x - 3)(x + 2)(x^2 + 3x - 9)$
(d) $(x - 3)(x + 2)(x^2 + 3x + 9)$

Q.56. The product of LCM and HCF of two numbers is 24. If the difference between the two numbers is 2, then find the numbers.

(a) 5 and 3 (b) 6 and 4
(c) 2 and 4 (d) 8 and 6

Q.57. The LCM of

$\frac{1}{6}, \frac{4}{15}, \frac{5}{9}, \frac{7}{27}, \frac{8}{3}$ is:

(a) $92\frac{1}{3}$ (b) $93\frac{2}{3}$ (c) $93\frac{1}{3}$ (d) $92\frac{2}{3}$

Q.58. The largest number of four digits that is exactly divisible by 17 and 36 is:

(a) 8568 (b) 9180 (c) 9792 (d) 7956

Q.59. Two numbers are in the ratio 3 : 4. The product of their HCF and LCM is 2700. The sum of the numbers is:

(a) 45 (b) 105 (c) 60 (d) 15

Q.60. Find the greatest number which when divides 261, 853 and 1221, leaves a remainder of 5 in each case.

(a) 18 (b) 17 (c) 16 (d) 19

Q.61. Find the LCM of $\frac{3}{2}, \frac{81}{16}$ and $\frac{9}{8}$.

(a) $\frac{111}{2}$ (b) $\frac{91}{2}$ (c) $\frac{81}{2}$ (d) $\frac{101}{2}$

Q.62. Find the greatest possible length (in metres) that can be used to exactly measure the lengths 6 m, 5 m 25 cm and 12 m 50 cm.

(a) 0.35 m (b) 0.90 m (c) 0.75 m (d) 0.25 m

Q.63. The least number which should be added to 1351 so that the sum is exactly divisible by 2, 4, 6 and 8 is:

(a) 13 (b) 11 (c) 15 (d) 17

Q.64. Let the HCF of m and n be 'a' and let $n = ab$, LCM of m and n is given by:

(a) ab (b) am (c) bm (d) mn

Q.65. The LCM and HCF of two numbers are 1105 and 5. If the LCM is 17 times the first number, then find the two numbers.

(a) 55 and 85 (b) 65 and 75
(c) 60 and 80 (d) 65 and 85

Q.66. If the HCF of 45 and 55 is expressible in the form of $55 \times 5 + 45m$, then what is the value of m?

(a) 5 (b) -6 (c) -5 (d) 6

Q.67. Find the least number which when divided by 4, 9, 12 and 15, leaves the remainder 3 in each case.

(a) 193 (b) 183 (c) 360 (d) 180

Q.68. Determine the LCM of two numbers if their HCF is 12 and their ratio is 13 : 15.

(a) 2450 (b) 1780 (c) 1890 (d) 2340

Q.69. Which of the following numbers leaves the remainder equal to the highest common factor of 6, 8 and 9, when divided by 6, 8 and 9?

(a) 506 (b) 575 (c) 291 (d) 433

Q.70. Find the least number which is exactly divisible by 20, 28, 34, 60 and 75.

(a) 34500 (b) 35900 (c) 35700 (d) 36220

Q.71. Which is the largest number that divides each of 1036, 1813 and 3885 without leaving any remainder?

(a) 259 (b) 111 (c) 333 (d) 37

Q.72. What is the LCM of 0.15, 0.18 and 0.45?

(a) 0.6 (b) 0.9 (c) 0.81 (d) 0.09

Q.73. What will be the least number which when doubled will be exactly divisible by 12, 14, 16 and 18?

(a) 636 (b) 226 (c) 428 (d) 504

Q.74. The LCM of two numbers is five times their HCF. If the product of the two numbers is 20480, then find their HCF and LCM, respectively.

(a) 64 and 320 (b) 56 and 280
(c) 48 and 240 (d) 46 and 230

Q.75. If the sum of two numbers is 60 and their HCF and LCM are 5 and 60, respectively, then the sum of the reciprocals of the numbers will be:

(a) $\frac{1}{4}$ (b) $\frac{1}{5}$ (c) $\frac{1}{11}$ (d) $\frac{1}{6}$

Q.76. The least number of five digits which is exactly divisible by 9, 12, 15, 25 and 27 is:

(a) 10250 (b) 10800 (c) 10600 (d) 10700

Q.77. What is the largest number that divides 627, 15630 and 3128 and leaves

remainders of 2, 5 and 3, respectively?

(a) 775 (b) 650 (c) 625 (d) 1225

Q.78. The HCF of three numbers is 57. If they are in the ratio of 4 : 5 : 6, then find the numbers.

(a) 228, 285, 342 (b) 236, 295, 354
(c) 240, 300, 360 (d) 232, 290, 348

Q.79. Three numbers are in the ratio 3 : 5 : 7. If their LCM is 2625, find their HCF.

(a) 100 (b) 125 (c) 50 (d) 25

Q.80. The largest three-digit number which gives the same remainder 2 when divided by 3, 5 and 9 is _____.

(a) 980 (b) 992 (c) 995 (d) 990

Q.81. The HCF of two numbers is 12. Which of the following can never be their LCM?

(a) 24 (b) 74 (c) 48 (d) 60

Q.82. The product of two numbers is 4107. If the HCF of these numbers is 37, then the greater number is:

(a) 111 (b) 74 (c) 185 (d) 37

Q.83. The LCM of the two numbers is 1440 and their HCF is 32. If the first number is 288, then find the second number.

(a) 150 (b) 145 (c) 180 (d) 160

Q.84. What is the greatest three-digit number which when divided by 4, 8 and 9, leaves no remainder?

(a) 924 (b) 963 (c) 932 (d) 936

Q.85. Find the smallest three digit number which when added to 5 is exactly divisible by 12, 14 and 18.

(a) 240 (b) 225 (c) 245 (d) 247

Q.86. The ratio of two numbers is 3 : 4. If the sum of these two numbers is 70, then what is their LCM?

(a) 120 (b) 96 (c) 108 (d) 84

Q.87. Find the least number divisible by 2, 3, 5, 6, 9 and 18, which is a perfect square.

(a) 900 (b) 400 (c) 144 (d) 3600

Q.88. The smallest six-digit number which is divisible by 12, 16, 24, 28 is:

(a) 100128 (b) 100190
(c) 100180 (d) 100160

Q.89. The HCF of two numbers is one-twentieth of their LCM. If one of the numbers is 96 and the difference of the LCM and the HCF is 456, then what is the other number?

(a) 48 (b) 120 (c) 144 (d) 72

Q.90. What is the HCF of two prime numbers X and Y?



(a) 1 (b) 2 (c) Y (d) X

Q.91. What is that least number that should be added to 478 so that the number obtained becomes exactly divisible by 5, 6 and 12 ?

(a) 62 (b) 12 (c) 2 (d) 52

Q.92. The product of two co-prime numbers is 483. Find the Least Common Multiple of both numbers.

(a) 483 (b) 21 (c) 1 (d) 23

Q.93. What is that least perfect square which is exactly divisible by each of 14, 84 and 28 ?

(a) 7056 (b) 3528 (c) 1764 (d) 441

Q.94. The greatest number which when divides 456 and 553 leaves the remainder as 6 and 3 respectively is:

(a) 5 (b) 50 (c) 10 (d) 30

Q.95. The least number which when divided by 14, 18 and 36 and leaves 1 as remainder in each case is:

(a) 250 (b) 252 (c) 253 (d) 251

Q.96. What will be the ratio of the Least Common Multiple and Highest Common Factor of 12, 24 and 36 ?

(a) 3 : 1 (b) 4 : 1 (c) 6 : 1 (d) 2 : 1

Q.97. What is the least number of students in a class such that they can make rows of 10, 15 and 20 and also the total number of students is a perfect square?

(a) 6400 (b) 1600 (c) 900 (d) 1800

Q.98. What is the HCF of 513 and 1017 ?

(a) 15 (b) 8 (c) 9 (d) 13

Q.99. The Least Common Multiple of three different numbers is 40. Which of the following cannot be their Highest Common Factor ?

(a) 20 (b) 25 (c) 5 (d) 10

Q.100. What is the smallest perfect square which is divisible by both 8 and 12 ?

(a) 100 (b) 144 (c) 121 (d) 196

Q.101. The product of the two numbers is 1500 and their HCF is 10. The number of such possible pairs is/are:

(a) 1 (b) 3 (c) 4 (d) 2

Q.102. If the highest common factor (HCF) of x and y is 15, then the HCF of $36x^2 - 81y^2$ and $81x^2 - 9y^2$ is divisible by

(a) 135 (b) 120 (c) 180 (d) 90

Q.103. Calculate the HCF of $\frac{12}{5}$, $\frac{14}{15}$ &

$$\frac{16}{17}$$

(a) $\frac{4}{255}$ (b) $\frac{3}{255}$ (c) $\frac{2}{255}$ (d) $\frac{1}{255}$

Q.104. Choose the correct statement from the following.

(a) HCF is the least common multiple of the given numbers.

(b) HCF of two or more numbers is the highest number which perfectly divides all the given numbers.

(c) HCF is also called the least common divisor.

(d) In prime factorisation method of HCF, the multiples of all the given numbers are listed.

Answer Key :-

1.(b)	2.(a)	3.(b)	4.(d)
5.(a)	6.(a)	7.(b)	8.(b)
9.(c)	10.(d)	11.(d)	12.(d)
13.(d)	14.(a)	15.(d)	16.(d)
17.(d)	18.(c)	19.(a)	20.(a)
21.(b)	22.(a)	23.(d)	24.(c)
25.(b)	26.(c)	27.(d)	28.(b)
29.(a)	30.(b)	31.(a)	32.(a)
33.(a)	34.(b)	35.(a)	36.(b)
37.(b)	38.(d)	39.(b)	40.(a)
41.(a)	42.(b)	43.(a)	44.(c)
45.(c)	46.(b)	47.(b)	48.(d)
49.(d)	50.(b)	51.(c)	52.(c)
53.(c)	54.(a)	55.(d)	56.(b)
57.(c)	58.(c)	59.(b)	60.(c)
61.(c)	62.(d)	63.(d)	64.(c)
65.(d)	66.(b)	67.(b)	68.(d)
69.(d)	70.(c)	71.(a)	72.(b)
73.(d)	74.(a)	75.(b)	76.(b)
77.(c)	78.(a)	79.(d)	80.(b)
81.(b)	82.(a)	83.(d)	84.(d)
85.(d)	86.(a)	87.(a)	88.(a)
89.(b)	90.(a)	91.(c)	92.(a)
93.(c)	94.(b)	95.(c)	96.(c)
97.(c)	98.(c)	99.(b)	100.(b)
101.(d)	102.(a)	103.(c)	104.(b)

Solutions :-**Sol.1.(b)** LCM of (3 and 5) = 15Two-digit multiples of 15: 15 to 90 (AP with $a = 15$, $d = 15$, $n = 6$).

$$\text{Sum} = \frac{n}{2}(a + l)$$

$$= \frac{6}{2}(15 + 90) = 315$$

Sol.2.(a) L.C.M (a, b) = $2^3 \cdot 3^2 \cdot 5^2 \cdot 7$ **Sol.3.(b)** According to the question,

$$x^2 - 8x + 15$$

$$x^2 - (5 + 3)x + 15$$

$$x(x - 5) - 3(x - 5)$$

$$(x - 5), (x - 3)$$

$$\text{And } x^2 - 5x + 6$$

$$x^2 - (3 + 2)x + 6$$

$$x(x - 3) - 2(x - 3)$$

$$(x - 3), (x - 2)$$

So the LCM will be $(x - 5), (x - 2), (x - 3)$ **Sol.4.(d)** According to the question,

$$1 \text{ m} = 100 \text{ cm}$$

$$\text{HCF of } 315, 500, 685 = 5$$

$$315 = 5 \times 3 \times 21$$

$$500 = 5 \times 5 \times 20$$

$$685 = 5 \times 137$$

So the greatest possible length be 5 cm

Sol.5.(a)

$$(x^6 + 1) = (x^2 + 1)(x^4 + 1 - x^2)$$

$$(x^4 - 1) = (x^2 + 1)(x^2 - 1)$$

$$\text{H.C.F} = (1 + x^2)$$

Sol.6.(a) L.C.M = 533

As we know that L.C.M of two prime numbers is always the product of the two prime numbers

So, numbers = 13 and 41

$$\text{Hence, } 4y - x = 13 \times 4 - 41 = 52 - 41 = 11$$

Sol.7.(b)

$$(x^3 - 8) = (x - 2)(x^2 + 2x + 4)$$

$$(x^3 - 6x^2 + 12x - 8) = (x - 2)^3$$

$$\text{and } (x^3 - 4x^2 + 4x) = x(x - 2)^2$$

HCF of $(x^3 - 8)$, $(x^3 - 6x^2 + 12x - 8)$ and

$$(x^3 - 4x^2 + 4x) = (x - 2)$$

Sol.8.(b) Required time = LCM of 50, 75 and 100 $\Rightarrow 300 = 5$ minutes**Sol.9.(c)**

HCF of 7000, 3850 and 12950

$$7000 = 350 \times 20$$

$$3850 = 350 \times 11$$

$$12950 = 350 \times 37$$

So the maximum length will be = 350 mm or 35 cm

Sol.10.(d)

Maximum capacity of container = HCF (850, 680) = 170 litres

Sol.11.(d)

(219 - 3), (365 - 5), (511 - 7) will be

exactly divisible by that two digit number
Required number = HCF of 216, 360, 504 = 72**Sol.12.(d)**

L.C.M of 5, 10, 15, 20 and 25 = 300 sec.

$$\text{Required number} = \frac{80 \times 60}{300} + 1$$

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$$\begin{array}{r} 6 \\ 420 \overline{) 2800} \\ \underline{-2520} \\ 280 \end{array}$$

On Subtracting, $420 - 280 = 140$

Now, Adding 140 to 2800

$$2800 + 140 = 2940$$

As per question, 1 is the remainder that we get in each case but 2940 is completely divisible by 2, 3, 4, 5, 6 and 7

Therefore,

$$\text{the number} = 2940 + 1 = 2941$$

Now, the sum of the digits is 16.

$$\text{Sol.35.(a)} \quad 15 = 3 \times 5$$

$$18 = 2 \times 3 \times 3; 20 = 2 \times 2 \times 5$$

$$27 = 3 \times 3 \times 3$$

LCM of 15, 18, 20 and 27

$$= 2 \times 2 \times 3 \times 3 \times 3 \times 5 = 540$$

$$\Rightarrow x \text{ must be } = 540k + 10$$

Where $540k + 10$ is multiple of 31

The condition gets satisfied when $k = 4$

$$\text{Required number} = 540(4) + 10 = 2170$$

$$\text{Nearest square to 2170} = 2209$$

$$\text{Required number} = 2209 - 2170 = 39$$

$$\text{Sol.36.(b)} \quad 12 = 2 \times 2 \times 3; 16 = 2 \times 2 \times 2 \times 2$$

$$18 = 2 \times 3 \times 3; 20 = 2 \times 2 \times 5$$

$$25 = 5 \times 5$$

$$\text{LCM of } 12, 16, 18, 20 \text{ and } 25 = 2 \times 2 \times 2 \times 2$$

$$\times 3 \times 3 \times 5 \times 5 = 3600$$

$$\Rightarrow x \text{ must be } = 3600k + 4$$

Where $3600k + 4$ is multiple of 7

The condition gets satisfied when $k = 5$

$$\text{Required number} = 3600(5) + 4 = 18004$$

$$\Rightarrow \text{digit at the thousands' place in } x = 8$$

Sol.37.(b) If the number 'a' and 'b' are divisible by a number 'n' then, $a + b$

and $a - b$ are also divisible by n.

Here the numbers are 7531, 7105 and 6892

the required number then becomes H.C.F of 7531 - y, 6892 - y and 7105 - y or the HCF of $(7531 - y) - (7105 - y)$ and $(7105 - y) - (6892 - y)$

$$(7531 - y) - (7105 - y) = 426$$

$$(7105 - y) - (6892 - y) = 213$$

$$213 = 3 \times 71 \Rightarrow 426 = 2 \times 3 \times 71$$

So, HCF of 426 and 213

$$x = 3 \times 71 = 213$$

The remainder =

$$\frac{7531}{213} = \frac{7105}{213} = \frac{6892}{213} = y = 76$$

$$\Rightarrow (x - y) = 213 - 76 = 137$$

Sol.38.(d) x will be the HCF of the differences of the three numbers.

$$6475 - 4984 = 1491$$

$$4984 - 4132 = 852$$

$$6475 - 4132 = 2343$$

$$1491 = 3 \times 7 \times 71$$

$$852 = 2 \times 2 \times 3 \times 71$$

$$2343 = 3 \times 11 \times 71$$

So,

$$\text{HCF of } 2343, 1491 \text{ and } 852 = 3 \times 71 = 213$$

$$\text{So, sum of digit} = 2 + 1 + 3 = 6$$

$$\text{Sol.39.(b)} \quad \text{LCM of } 6, 9 \text{ and } 15 = 90$$

$$\text{So, least square number} = 90 \times 10 = 900$$

Sol.40.(a)

$$\text{LCM of } 3, 5, 12, 15 = 60 \Rightarrow 2 \times 3 \times 10$$

By checking the options one by one, the option(a) satisfies

Divisibility rule of 3:

If the sum of all digits of a number is divisible by 3, then the number itself is divisible by 3

$$\text{Sol.41.(a)} \quad \text{LCM of } 10 \text{ and } 15 = 30 \text{ minute}$$

$$\text{Sol.42.(b)} \quad \text{LCM of } (6, 9, 10) = 90$$

$$\text{Reqd. number} = 90 + 4 = 94$$

$$\text{Sol.43.(a)} \quad \text{LCM of } (3, 4, 5, 6, 8) = 2 \times 2 \times 3 \times 5 = 120$$

Now the smallest number that leaves a remainder of 4 when divided by each

$$\text{i.e. } 120 + 4 = 124$$

Sol.44.(c)

For same quantity of milk = HCF of (75, 45, 30) = 15 liter

$$\text{Trays for cow milk} = \frac{75}{15} = 5$$

$$\text{Trays for toned milk} = \frac{45}{15} = 3$$

$$\text{Trays for full cream milk} = \frac{30}{15} = 2$$

$$\text{Total trays} = 5 + 3 + 2 = 10$$

The least number of trays required to place all the milk is 10.

Sol.45.(c)

Let three numbers are $5x, 7x, 9x$

According to question,

$$(5 \times 7 \times 9)x = 1260$$

$$\Rightarrow 315x = 1260 \Rightarrow x = 4$$

$$\therefore \text{Three numbers are } 20, 28, 36$$

$$\text{Required difference} = 36 - 20 = 16$$

$$\text{Sol.46.(b)} \quad \text{LCM}(4, 6, 8) = 24$$

It means that in every 24 minute red light blinks

Now, 3 hours = 180 minute

So, in the interval of 3 hours red light

$$\text{blinks} = \frac{180}{24} = 7 \text{ times}$$

(consider integer value only)

Sol.47.(b)

Amita, Amrita and Tosita went office on 2nd, 3rd and 6th day respectively

$$\text{LCM}(2, 3, 6) = 6$$

So, on every 6th day all three attend the office

Now, total days in last two months

$$= 30 + 31 = 61 \text{ days}$$

Hence, no. of times they attend together

$$\text{the office in last two months} = \frac{61}{6} = 10$$

times (only integer value considered)

$$\text{Sol.48.(d)} \quad \text{LCM}(20, 25, 35, 40)$$

$$= 2^3 \times 5 \times 5 \times 7 = 1400$$

Now, in every case remainder is 6 less than the divisor

$$\text{So, required no.} = 1400 - 6 = 1394$$

$$\text{Sol.49.(d)} \quad \text{Given, } x = a^2b^3 \text{ and } y = a^3b$$

$$\text{HCF}(x, y) = a^2b$$

Sol.50.(b) According to question,

$$\text{Given: } x = 2k, y = 3m, z = 4n.$$

$$\text{Then, } 6x = 12k, 4y = 12m, 3z = 12n.$$

All are multiples of 12.

$$\text{HCF}(6x, 4y, 3z) = 12.$$

Hence, the HCF(6x, 4y, 3z) is at least 12.

$$\text{Sol.51.(c)} \quad \text{Given, } 3^3 \times 11^2, 2^4 \times 3^2 \times 5^3 \times 11^3 \text{ and } 3 \times 5 \times 7 \times 11$$

$$\text{HCF} = 3 \times 11 = 33$$

Sol.52.(c)

Let the number be $7x$ and $9x$ respectively,

According to question,

$$\Rightarrow 7 \times 9 \times x = 630 \Rightarrow x = 10$$

So, the number will be 70 and 90.

$$\text{Sol.53.(c)} \quad \text{HCF of } (156 \text{ and } 96) = 12$$

According to question,

$$\Rightarrow 156 \times 12 + 96A = 12$$

$$\Rightarrow 1872 + 96A = 12$$

$$\Rightarrow A = \frac{12 - 1872}{96} = \frac{-1860}{96} = \frac{-155}{8}$$

$$= -19.375$$

$$\text{Sol.54.(a)} \quad \text{Numbers} = 1.75, 5.6, 7$$

$$= \frac{175}{100}, \frac{560}{100}, \frac{700}{100}$$

$$\text{HCF} = \frac{\text{HCF of } (175, 560, 700)}{\text{LCM of } (100, 100, 100)} = \frac{35}{100}$$

$$= 0.35$$

Sol.55.(d)

$$\text{Factor of } (x^3 - 27) = (x - 3)(x^2 + 9 + 3x)$$

$$\text{Factor of } (x^2 - x - 6) = (x + 2)(x - 3)$$

$$\text{Factor of } (x + 2) = (x + 2)$$

$$\text{LCM of } (x^3 - 27), (x^2 - x - 6) \text{ and } (x + 2)$$

$$= (x - 3)(x + 2)(x^2 + 9 + 3x)$$

Sol.56.(b) Product of two numbers

$$= \text{LCM} \times \text{HCF} = 24$$

From the options we can see that only option (b) has numbers which can give 24 as product.

Hence, two numbers are 6 and 4.

$$\text{Sol.57.(c)} \quad \text{L.C.M. of } 1, 4, 5, 7, 8 = 280$$

$$\text{H.C.F. of } 6, 15, 9, 27, 3 = 3$$

L.C.M. of fractions

$$= \frac{\text{L.C.M. of numerator}}{\text{H.C.F. of denominator}} = \frac{280}{3} = 93\frac{1}{3}$$

$$\text{Sol.58.(c)} \quad \text{L.C.M of } 17 \text{ and } 36 = 612$$

Largest number of 4-digit = 9999



When we divide 9999 by 612, we get 207 as a remainder.

Required number = $9999 - 207 = 9792$

Sol.59.(b) Let the numbers are $3x$ and $4x$
According to question,
 $\Rightarrow 3x \times 4x = 2700 \Rightarrow x = 15$
So, the sum of the numbers
 $= 15 \times 3 + 15 \times 4 = 45 + 60 = 105$

Short Trick :- $(3x + 4x) = 7x$

Hence, the sum of the numbers must be a multiple of 7.

Now, from options, 105 is the required answer.

Sol.60.(c) According to the question,
 $261 - 5 = 256 \Rightarrow 16 \times 16$
 $853 - 5 = 848 \Rightarrow 16 \times 53$
 $1221 - 5 = 1216 \Rightarrow 16 \times 76$
So the HCF of 256, 848, 1216 will be 16
So the required number will be 16

Sol.61.(c) According to the question,
LCM of $\frac{3}{2}, \frac{81}{16}, \frac{9}{8} = \frac{LCM}{HCF} = \frac{81}{2}$

Sol.62.(d) According to the question,
1 m = 100 cm
HCF of (600, 525, 1250) = 25 cm or 0.25 m
So the greatest possible length will be 0.25 m

Sol.63.(d) L.C.M of 2, 4, 6 and 8 = 24
When we divide 1351 by 24 we get 7 as remainder.
So, the required number = $24 - 7 = 17$

Sol.64.(c) H.C.F of m and $n = a$
Given, $n = ab$
L.C.M = $\frac{\text{product of numbers}}{H.C.F}$
 $= \frac{m \times ab}{a} = mb$

Sol.65.(d) According to the question,
 $1105 = 17 \times 1^{\text{st}} \text{ number}$
 $\Rightarrow 1^{\text{st}} \text{ number} = 65$
Formula:- $1^{\text{st}} \text{ no.} \times 2^{\text{nd}} \text{ no.} = \text{LCM} \times \text{HCF}$
 $65 \times 2^{\text{nd}} \text{ no.} = 1105 \times 5$
 $2^{\text{nd}} \text{ no.} = 85$
So the no. will be 65 and 85

Sol.66.(b) $\text{HCF}(45, 55) = 5$
According to the question,
 $5 = 55 \times 5 + 45m$ -----(1)
By go through the options one by one
Value $m = -6$ satisfied the equations
i.e. $5 = 55 \times 5 + 45 \times (-6)$
 $\Rightarrow 5 = 275 - 270 = 5$

Sol.67.(b) $\text{LCM}(4, 9, 12, 15) = 180$
hence, least number = $180 + 3 = 183$

Sol.68.(d)
Let numbers be $13x$ and $15x$ respectively,
Then, $\text{HCF}(13x, 15x) = x$
According to the question,

$x = 12$
So, Numbers = 156 and 180
Hence, $\text{LCM}(156, 180) = 2340$

Sol.69.(d) Remainder = $\text{HCF}(6, 8, 9) = 1$
And $\text{LCM}(6, 8, 9) = 72$
So, number should be = $72k + 1$
After putting $k = 6$ we get;
Number = $72 \times 6 + 1 = 433$

Sol.70.(c)
L.C.M of 20, 28, 34, 60 and 75 = 35700
So, the required least number = 35700

Sol.71.(a)
H.C.F of 1036, 1813 and 3885 = 259
Hence, the required largest number = 259

Sol.72.(b) According to the question,
 $\text{LCM of } \frac{15}{100}, \frac{18}{100}, \frac{45}{100} = \frac{90}{100} = 0.9$

Sol.73.(d) According to the question,
LCM of 12, 14, 16 and 18 be 1008
So the 504 is the least number when it doubles will be exactly divisible by the 12, 14, 16, and 18.

Sol.74.(a) Given,
 $\text{LCM} = 5 \times \text{HCF}$
Let numbers be "a" and "b"
 $a \times b = 20480$
 $\text{HCF} \times \text{LCM} = a \times b$
 $\text{HCF} \times 5 \times \text{HCF} = 20480$
 $\text{HCF}^2 = \frac{20480}{5} = 4096 \Rightarrow \text{HCF} = 64$
 $\text{LCM} = 5 \times \text{HCF} = 5 \times 64 = 320$

Sol.75.(b) Let numbers be a and b
 $a \times b = \text{HCF} \times \text{LCM}$
 $a \times b = 5 \times 60$ -----(1)
 $a + b = 60$ -----(2)
Now,
 $\frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab} = \frac{60}{5 \times 60} = \frac{1}{5}$

Sol.76.(b) According to the question,
LCM of 9, 12, 15, 25 and 27 will be 2700
So the smallest 5 digit number which exactly divisible by these numbers
 $= 2700 \times 4 = 10800$

Sol.77.(c) According to the question,
1st number = $627 - 2 = 625$
2nd number = $15630 - 5 = 15625$
3rd number = $3128 - 3 = 3125$
 $\text{HCF of } 625, 15625 \text{ and } 3125 = 625$
So the required number will be 625.

Sol.78.(a) According to the question,
No. will be = $4 \times 57, 5 \times 57, 6 \times 57$
 $= 228, 285, 342$

Sol.79.(d) Let three numbers be $3x, 5x$ and $7x$ respectively,
Then, $\text{LCM}(3x, 5x, 7x) = 105x$
 $105x = 2625 \Rightarrow x = 25$

numbers = $(3 \times 25), (5 \times 25), (7 \times 25)$
So, HCF = 25

Sol.80.(b) $\text{LCM}(3, 5, 9) = 45$
Now, remainder $\frac{999}{45} = 9$
So, required largest three digit number
 $= (999 - 9) + 2 = 992$

Sol.81.(b) According to the question,
If the HCF of any 2 numbers are x then the LCM of their will be the multiple of x .
So in option (b) 74 is not the multiple of 12 so option (b) is the right answer

Sol.82.(a)
Let the numbers be $37x$ and $37y$
According to the question,
 $(37 \times x) \times (37 \times y) = 4107$
 $xy = 3$ {where x and y be co-prime}
So, possible value of x and y be 1 and 3
Hence, the greater number = $37 \times 3 = 111$

Sol.83.(d)
 $\text{LCM} \times \text{HCF} = 1^{\text{st}} \text{ number} \times 2^{\text{nd}} \text{ number}$
 $1440 \times 32 = 288 \times 2^{\text{nd}} \text{ number}$
2nd number = 160

Sol.84.(d) LCM of 4, 8, 9 = 72
Greatest three digit number = 999
 $\frac{999}{72} = \text{Rem}(63)$
Required number = $999 - 63 = 936$

Sol.85.(d) The smallest number which is completely divisible by 12, 14 and 18 will be the LCM of these numbers.
So the LCM of 12, 14 and 18 = 252
Required number = $252 - 5 = 247$

Sol.86.(a) Let the numbers are $3x$ and $4x$
According to the question,
 $3x + 4x = 70 \Rightarrow x = 10$
Numbers 30 and 40
LCM of 30 and 40 = 120

Sol.87.(a) L.C.M of 2, 3, 5, 6, 9 and 18 = 90
 $90 = 2 \times 3 \times 3 \times 5$
As 2 and 5 don't have any pair
So,
the required number = $90 \times 2 \times 5 = 900$

Sol.88.(a)
L.C.M. (12, 16, 24, 28) = $3 \times 4 \times 4 \times 7$
 $= 336$
Smallest 6 digit no. = 100000
Rem. $(\frac{100000}{336}) = 208$
So, required smallest 6 digit no.
 $= 100000 + (336 - 208) = 100128$

Sol.89.(b) Let,
L.C.M. be 20 units then H.C.F. be 1 unit.
According to the question,
Difference (19 units) = $456 \Rightarrow 1 \text{ unit} = 24$
Now, L.C.M. $\times$ H.C.F. = $1^{\text{st}} \text{ no.} \times 2^{\text{nd}} \text{ no.}$
 $(20 \times 24) \times (1 \times 24) = 96 \times 2^{\text{nd}} \text{ no.}$

$$\Rightarrow 2\text{nd no.} = \frac{(20 \times 24) \times (1 \times 24)}{96} = 120$$

Sol.90.(a) The HCF of two prime numbers is always 1.

Sol.91.(c) LCM (5, 6, 12) = 60

$$\text{Remainder} = \frac{478}{60} = 58$$

Hence,

$$\text{Required least number} = 60 - 58 = 2$$

Sol.92.(a)

Product of two co - prime number = 483

HCF of co - prime numbers always be 1

So, least common multiple of both numbers = $1 \times 483 = 483$

Sol.93.(c) $14 = 2 \times 7$, $84 = 2^2 \times 3 \times 7$

$$28 = 2^2 \times 7$$

Required perfect square no.

$$= 2^2 \times 3^2 \times 7^2 = 1764$$

Sol.94.(b) According to the question,

Number that completely divisible,

$$456 - 6 = 450 \Rightarrow 553 - 3 = 550$$

According to the option 50 is the largest number that divides 450 and 550 completely.

Sol.95.(c) The number which is divided by 14, 18 and 36 $\Rightarrow$ LCM(14, 18, 36) = 252

Remainder = 1{given}

So, Required number = 253

Sol.96.(c) Least common multiple of 12, 24 and 36 = 72

Highest common multiple of

12, 24 and 36 = 12

Required ratio = $72 : 12 = 6 : 1$

Sol.97.(c) According to question,

LCM (10, 15, 20) = 60

From the given options,

option (c) $(60 \times 15) = 900$,

is the correct answer.

Sol.98.(c) $513 = 3 \times 3 \times 57$

$$1017 = 3 \times 3 \times 113$$

So, H.C.F.(513, 1017) = $3 \times 3 = 9$

Sol.99.(b) Factor of 40 = $2 \times 2 \times 2 \times 5$

By the option 25 never be the HCF

Sol.100.(b) Smallest number that is divisible by 8 and 12 will be 24

Required number = $24 \times 6 = 144$

Sol.101.(d)

Let the numbers be $10x$ and $10y$

$$\text{Then, } 10x \times 10y = 1500 \Rightarrow xy = 15$$

Possible value of x and y are

$$\Rightarrow (1, 15) \text{ and } (3, 5)$$

Sol.102.(a) Let, the no. are $15a$ and $15b$.

$$36x^2 - 81y^2 = 9(2x - 3y)(2x + 3y)$$

$$= 9(2 \times 15a - 3 \times 15b)(2 \times 15a + 3 \times 15b)$$

$$= 9 \times 15\{(2a - 3b)(2a + 3b)\}$$

$$\text{And, } 81x^2 - 9y^2 = 9(3x - y)(3x + y)$$

$$= 9(3 \times 15a - 15b)(3 \times 15a + 15b)$$

$$= 9 \times 15\{(3a - b)(3a + b)\}$$

Their H.C.F. = 135

Sol.103.(c) H.C.F. $(\frac{12}{5}, \frac{14}{15}, \frac{16}{17})$

$$= \frac{H.C.F.(12, 14, 16)}{L.C.M.(5, 15, 17)} = \frac{2}{255}$$

Sol.104.(b) H.C.F of two or more numbers is the highest number which exactly divides all the given numbers.

Example:- H.C.F of 4 and 6 = 2

2 exactly divides 4 and 6.

